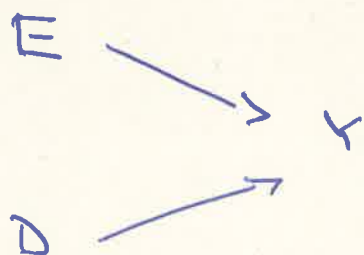


Association in the case of common effects

- E: Gene of interest
- D: Mental disorder
- Y: Ecstasy consumption

- (Unknown) situation



That is: $RR_{ED} = 1$

$$RR_{EY} > 1$$

$$RR_{DY} > 1$$

Objective: $\hat{RR}_{ED}$

- What may happen if the sample consists of ecstasy users?

If consumption is not due to E, it is more probable that it is because of D, that is,

$$\hat{P}(D|E) < \hat{P}(D|\bar{E}) \leadsto \hat{RR}_{ED} < 1 = RR_{ED}$$

- Extreme case: E and D are the only risk factors of $Y=1$

Hence, if all sample units have the same characteristic ($Y=1$), then $\hat{RR}_{ED} = \frac{\hat{P}(D|E)}{\hat{P}(D|\bar{E})} = \hat{P}(D|E) < 1$